

Tomato Is Life: A Prototype for Sorting Tomatoes Based on Size, Colors, and Quality

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Abstract—This study presents an automated tomato sorting system designed to classify tomatoes based on size (small and large), color (red and green), and bad quality. The research addresses the challenges and drawbacks of traditional manual sorting, which is labor-intensive, prone to errors, and causes inconsistencies. The objective of this study is to develop an IoT model with computer vision techniques integrated with a machine learning algorithm and measure the segregation and time to complete the sorting. The system utilizes a web camera, conveyor belt, breadboard, power supply, and servo motors as the main components of the system. Images of tomatoes are captured in real-time and processed using Roboflow's Vision Transformer (ViT) model, which classifies them according to predefined parameters. Lastly, the classification results determine the movement of servo motors, which direct the tomatoes into designated containers. The findings reveal that the system achieves a high accuracy of more than 70% in sorting tomatoes in their correct bins with an average sorting time of 28 seconds per tomato. The study concludes that automated sorting systems using machine learning can provide a scalable and reliable solution for agricultural industries, minimize labor costs, and improve post-harvest management.

Keywords—Agriculture, Automated Sorting, Computer Vision, Machine Learning, Tomato Sorting

I. INTRODUCTION

Tomato (*Solanum lycopersicum*) is one of the most widely cultivated fruit crops in the Philippines due to its versatility in culinary applications and high nutritional value. As a warm-season crop, it thrives in various soil types and climatic conditions [1]. The Philippines, being an agricultural country, relies heavily on farming, particularly crop cultivation, which serves as a primary livelihood for many Filipinos [2]. However, traditional farming methods require significant human effort and time, making agriculture labor-intensive and inefficient.

Tomatoes are among the most profitable crops both locally and globally due to their high market demand and economic viability. Ensuring the delivery of high-quality tomatoes is essential for maintaining customer satisfaction, strengthening market competitiveness, and increasing profitability [3]. Despite its economic potential, the tomato industry faces challenges in sorting processes, often resulting

in inefficiencies, increased labor costs, and post-harvest losses.

Currently, tomato sorting is primarily performed manually, which presents several drawbacks, including high labor costs, worker fatigue, inconsistencies, and low accuracy [4]. Additionally, relying solely on human judgment for classification increases the risk of sorting errors, leading to decreased product quality and economic losses for farmers [5]. These inefficiencies highlight the need for technological interventions to improve sorting precision and efficiency.

The Internet of Things (IoT) offers promising solutions for optimizing agricultural processes. Computer vision, in particular, enables the objective analysis of tomatoes based on predefined criteria such as color, size, and quality. Therefore, this study aims to develop a Smart Sorting Solution that integrates IoT to enhance sorting accuracy, reduce labor dependency, and minimize product wastage. By modernizing traditional sorting methods, this research seeks to address existing challenges in tomato classification, ultimately improving efficiency and quality control in the industry.

II. RELATED STUDIES

A. Image Processing and Computer Vision in Tomato Sorting

Phishing scams are fraudulent emails seeking to obtain The use of image processing and machine vision has been effective in assessing tomato quality and automating the sorting process. A study by Zuo [6] highlighted the advantages of automation in reducing manual labor while ensuring consistency in classification. However, challenges such as sensitivity to lighting variations and the need for advanced algorithms for real-time processing were identified. To address these issues, the study suggested incorporating deep learning models and improved sensor technologies.

A neural network-based tomato sorting system was developed using a conveyor belt, camera, and microcontroller, demonstrating a 96.47% accuracy rate in classifying tomato quality. Their findings emphasized the effectiveness of image processing in



agricultural automation, proving that computer vision can significantly enhance sorting precision [7].

A tomato sorting system utilizing the HSV color model for color detection, object inner points for size measurement, and a load cell sensor for weight classification achieved 100% accuracy in color detection and 95% accuracy in weight measurement. However, the system developed by Sari et al. [8] struggled with size classification, achieving only 5% accuracy. These results suggest that further improvements in size detection algorithms are necessary to enhance overall classification performance.

B. Machine Learning and Deep Learning in Tomato Classification

Machine learning techniques have gained significant traction in tomato classification due to their ability to process large datasets, recognize patterns, and enhance classification accuracy over time. Various models, including Artificial Neural Networks (ANNs), Support Vector Machines (SVMs), and Convolutional Neural Networks (CNNs), have been utilized to automate the grading process, improving efficiency in agricultural applications.

An automated tomato grading and classification system was developed by Behera et al. [9], utilizing digital imaging and MATLAB for preprocessing. The system classified tomatoes based on three size categories (small, medium, and large) and three varieties (cherry, classic, and cylindrical). The correlation coefficients (R^2 values) of 0.98 for length and 0.97 for width demonstrated a strong relationship between predicted and actual tomato measurements. These results indicate the high precision of the system in measuring tomato dimensions and classifying them accordingly.

The implementation of Convolutional Neural Networks (CNNs) has significantly improved classification accuracy in tomato sorting. A study conducted by Abekoon et al. [10] developed a CNN-based classification model that achieved 99% accuracy in both training and validation phases for binary classification (ripe and unripe tomatoes). Furthermore, a multi-class classification model attained 99% training accuracy and 98% validation accuracy, demonstrating its robustness in identifying different tomato categories. The study also examined the impact of ripeness on post-harvest longevity, concluding that red tomatoes exhibited the longest shelf life due to their optimal physiological and chemical properties.

Another approach involved integrating K-Means clustering for segmentation with SVM and ANN models for classification by Pati et al. [11], effectively distinguishing fresh tomatoes from defective ones. The study demonstrated that K-Means clustering improved image segmentation, allowing the machine learning models to more accurately differentiate between fresh and defective tomatoes. Despite the system's high performance, researchers recommended further refinements in feature selection to enhance classification accuracy, particularly in distinguishing minor defects from natural imperfections.

Artificial Neural Networks (ANNs) have also been employed in tomato classification by leveraging texture, size, and color features. Girma et al. [12] developed an ANN-based model that successfully classified tomatoes with high accuracy. The dataset comprised 180 images collected from multiple agricultural regions, ensuring diversity in tomato samples. MATLAB was used for preprocessing and feature extraction, contributing to the model's reliability in identifying different tomato characteristics.

Meanwhile, a CNN-based tomato grading system further advanced classification performance by implementing transfer learning to detect surface defects such as rotting and swelling. Tee et al. [13] reported a 97.61% accuracy rate in defect detection, highlighting the potential of deep learning models in identifying various tomato imperfections. Additionally, an OpenCV-based size measurement algorithm was developed to enhance sorting accuracy. The system achieved a Root Mean Square Error (RMSE) of 0.258 mm in size classification, indicating high precision in determining tomato dimensions. These findings suggest that integrating machine learning with computer vision significantly improves tomato classification and defect detection.

C. Hybrid and Alternative Approaches in Tomato Sorting

In addition to machine learning techniques, hybrid approaches combining different methodologies have been explored to improve tomato sorting accuracy. A fuzzy logic and machine vision-based sorting system demonstrated enhanced classification performance by incorporating uncertainty handling mechanisms. Unlike conventional machine learning models, fuzzy logic allows for more flexible decision-making by considering varying degrees of ripeness and defect severity. The study, conducted using MATLAB, showed promising results; however, scalability remained a challenge for large-scale applications. The researchers suggested further optimization of fuzzy-based sorting algorithms to facilitate commercial implementation [14].

An alternative method combining K-Means clustering and pixel area subtraction was introduced by Yong et al. [15] to improve classification accuracy. The system incorporated a conveyor belt, cameras, and a controlled lighting setup to enhance image acquisition quality. Experimental results demonstrated that the system achieved 90.00% accuracy in defect detection, 83.33% sorting accuracy, and an improved 95.24% detection coverage for damaged tomatoes. These findings underscore the importance of controlled lighting conditions in maximizing the performance of machine vision-based sorting systems. The researchers emphasized that optimizing illumination and background settings could further reduce misclassification errors, particularly in distinguishing similar-colored tomatoes.

III. METHODOLOGY

A. Dataset

The dataset used for training and testing was

collected from market-purchased tomatoes to ensure a diverse representation of real-world variations. A total of 11,398 high-resolution images were captured from eight different angles under controlled lighting conditions, maintaining consistency and reducing the impact of shadows and reflections. These images were then categorized to facilitate effective model training. As shown in Table I, the dataset is distributed across different classes. Table II outlines the classification of tomatoes, categorizing them into three distinct labels: Red, Green, and Bad. Meanwhile, Table III groups tomatoes into different categories based on predefined size ranges.

TABLE I. DATASET DISTRIBUTION

Category Name	Number of Images
Red	3,438
Green	3,473
Bad	4,487

TABLE II. TOMATO COLOR CLASSIFICATION

Class Name	Pictures
Green (both green and yellow)	
	
Red (both red and orange)	
	
Bad	



TABLE III. TOMATO SIZE CLASSIFICATION

Class Name	Picture	Max. Diameter	Max. Circumference
Small		< 5 cm	≈ 15 cm
Large		> 5 cm	≈ 40 cm

B. Computer Vision Model

This study undertook two tasks: one is classification (identifying what type of tomato), and the other is object detection (what the camera detects). Pre-trained models were selected from the Roboflow Platform, a cloud-based computer vision tool.

C. Size Detection

To determine the size of a tomato, a bounding box is created around it. The bounding box measures the width of the tomato in centimeters using a floating-point decimal function. The formula used for computing the real-world width from the bounding box width in pixels is:

$$\text{size} = (\text{bbpx} \div 225) * 15 \quad (1)$$

Formula 1 converts the bounding box width from pixels to centimeters. The bbpx is the associated bounding box measured in pixels, 225 is the set camera frame after calibration, and 15 is the referencing real-world size of a small tomato. After repeated trial, the set threshold for small tomato is 6.55, otherwise, it will be sorted as large tomato.

D. Materials and Cost

This research built a prototype using lightweight and alternative materials as illustrated in Table IV amounting to 12,843.00 pesos.

TABLE IV. MATERIALS AND COST IN BUILDING THE PROTOTYPE

Material	Quantity	Cost
Steel Foil	1 piece by roll	₱ 432.00
Plywood	1	₱ 880.00
Nail	50	₱ 5.00
Wood Screw	100	₱ 130.00
Jumper Wire - Male – Male - Female – Female - Male – Female	by set	₱ 190.00
Dc – Dc Buck Step Down Converter - Adjustable 5 Volts Fixed	2 pcs adjustable 1 pc 5 volt fixed	₱ 100.00 ₱ 60.00
Ac/dc 12 Volts 5 Ampere	1 pc	₱ 295.00
16 Channel 12-Bit PWM/Servo Driver-I2C Interface PCA9685 Module	1 pc	₱ 176.00
Raspberry Pi 4b	1 pc	₱ 3,750.00
Raspberry Pi 4 Case With Micro Fan	1pc	₱ 536.00
Web Camera 720p	1pc	₱ 1895.00
Servo Motor - Sg 92 - Sg 90	9 pcs overall all	sg 92- ₱ 160.00 sg 90 – ₱ 105.00
Bred Board	1 pc	₱ 85.00
Stepper Motor	1 pc	₱ 110.00
Stepper Motor Driver	1 pc	included with the stepper motor
Wire	1 pc 1 meter	₱ 35.00
Flash Light (LED Cube Light) Rechargeable	1 pc	₱ 160.00
Embossed Lnoli	1 pc 1 meter	₱ 99.00
No Nails	2 pc	₱ 85.00
Prekit Arduino	1 pc	₱ 1,179.00
Bearing	8 pc	₱ 55.00
Pipe 3 Inch	1pc	1,248.00
Pipe 2 Inch	1 pc	1,000.00
Steel Rod	4 pc	₱ 65.00
Washer	6 pc	₱ 8.00

TABLE V. VALIDATION ACCURACY BETWEEN ROBOFLOW MODELS

Model	Validation Accuracy
ViT	99.5%
ResNet	98.3%



Fig. 1. Prototype Model of the Tomato Sorting

E. Statistical Tools

This study employs statistical tools to evaluate the performance, accuracy, and efficiency of the tomato sorter. The key metrics used include overall accuracy both in size and colors, and average sorting time per tomato, as defined in Formula 2 and 3, respectively.

$$\text{accuracy} = \frac{(\text{correct prediction})}{\text{total prediction}} \times 100 \quad (2)$$

$$\text{time} = \frac{\text{sorting} (\text{end time-start time in mins}) \times 60 \text{ secs}}{\text{no. of tomatoes sorted}} \quad (3)$$

IV. RESULTS AND DISCUSSION

A. Evaluation of Roboflow Models

After annotation, the training model was initiated, and customization options such as source image selection, train/test split (default), and preprocessing (augmentation: grayscale and saturation percentage) were applied. In Table V, a comparison between the Vision Transformer (ViT) and Residual Network (ResNet) models is presented based on their validation accuracy. The results indicate that the ViT model achieved a validation accuracy of 99.5%, whereas the ResNet model attained 98.3%. This comparison highlights that ViT performed better in classifying tomatoes based on their color and quality.

TABLE VI. INPUT TO THE TOMATO SORTER BEFORE THE EXPERIMENT

Tomato (pcs)	Size		Color			Start Time
	Small	Large	Red	Green	Bad	

10	3	4	3	4	3	09:27 AM
10	4	2	4	2	4	09:42 AM
10	5	2	4	3	3	10:18 AM

TABLE VII. OUTPUT OF THE TOMATO SORTER AFTER THE EXPERIMENT

Tomato (pcs)	Size		Color			End Time
	Small	Large	Red	Green	Bad	
10	3	2	3	2	3	09:32 AM
10	3	2	3	2	3	09:46 AM
10	4	2	4	2	1	10:23 AM

TABLE VIII. ACCURACY SCORE IN TERMS OF SIZE, COLOR, AND QUALITY

Criteria in Sorting	Accuracy
Small	85.0%
Large	83.33%
Red	91.67%
Green	72.33%
Bad	69.33%

B. Experiment Conducted in the Tomato Sorting Machine

To assess the performance of the developed tomato sorting machine as assembled in Figure 1, a series of experiments were conducted to measure its accuracy in classifying tomatoes based on size, color, and quality. The experiment involved processing three batches of tomatoes, each containing ten samples. The input parameters recorded included the actual number of small and large tomatoes, their colors (red, green, or bad), and the start time of sorting. The output results in Table VII were then compared against the actual values in Table VI to determine the accuracy of classification.

C. Evaluation of the Tomato Sorting Machine

In Table VIII, the sorting system demonstrated varying levels of accuracy across different criteria, with the highest accuracy observed in identifying red tomatoes at 91.67%, indicating a strong performance in distinguishing them from other categories. Sorting small tomatoes also showed a high accuracy of 85%, while large tomatoes followed closely at 83.33%, suggesting that the system effectively differentiates based on size. However, the accuracy for green tomatoes was lower at 72.33%, indicating some difficulty in correctly classifying them. The lowest accuracy was recorded for bad tomatoes at 69.33%, suggesting that the system struggled to accurately identify defective or rotten tomatoes, likely due to the single-sided camera angle, which may have limited the system's ability to detect defects on all sides of the tomatoes.

Additionally, the results indicate that the machine processes each tomato in approximately 28 seconds,

demonstrating a reasonable sorting time for real-world applications. This efficiency is crucial in ensuring the system's practicality and usability in large-scale operations.

Overall, the average accuracy of the sorting system was calculated to be 80.33%, demonstrating a generally reliable performance. However, the lower accuracy for green and bad tomatoes suggests areas for improvement, particularly in enhancing the system's ability to distinguish between these categories. Further optimizations, such as refining image recognition algorithms or improving sensor accuracy, may help increase the overall sorting efficiency and reliability, making the system more effective for agricultural and industrial applications.

V. CONCLUSION

This study successfully developed an automated tomato sorting system that classifies tomatoes based on size, color, and quality using computer vision and machine learning. By integrating IoT technology with a Vision Transformer (ViT) model, the system achieved an average sorting accuracy of 80% and processed each tomato in approximately 28 seconds. These findings demonstrate the potential of automation in reducing labor dependency, improving classification precision, and minimizing post-harvest losses in the agricultural sector.

The prototype provides a scalable and cost-effective solution for modernizing traditional sorting methods, making it a valuable tool for farmers and agribusinesses. However, certain limitations, such as occasional misclassification and variations in accuracy, highlight areas for improvement. Future work may explore enhancing model performance with larger datasets, incorporating multi-spectral imaging for better defect detection, and optimizing sorting speed through improved hardware integration. Additionally, the system could be expanded to support real-time monitoring and data analytics, enabling farmers to make informed decisions about crop management and market distribution.

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